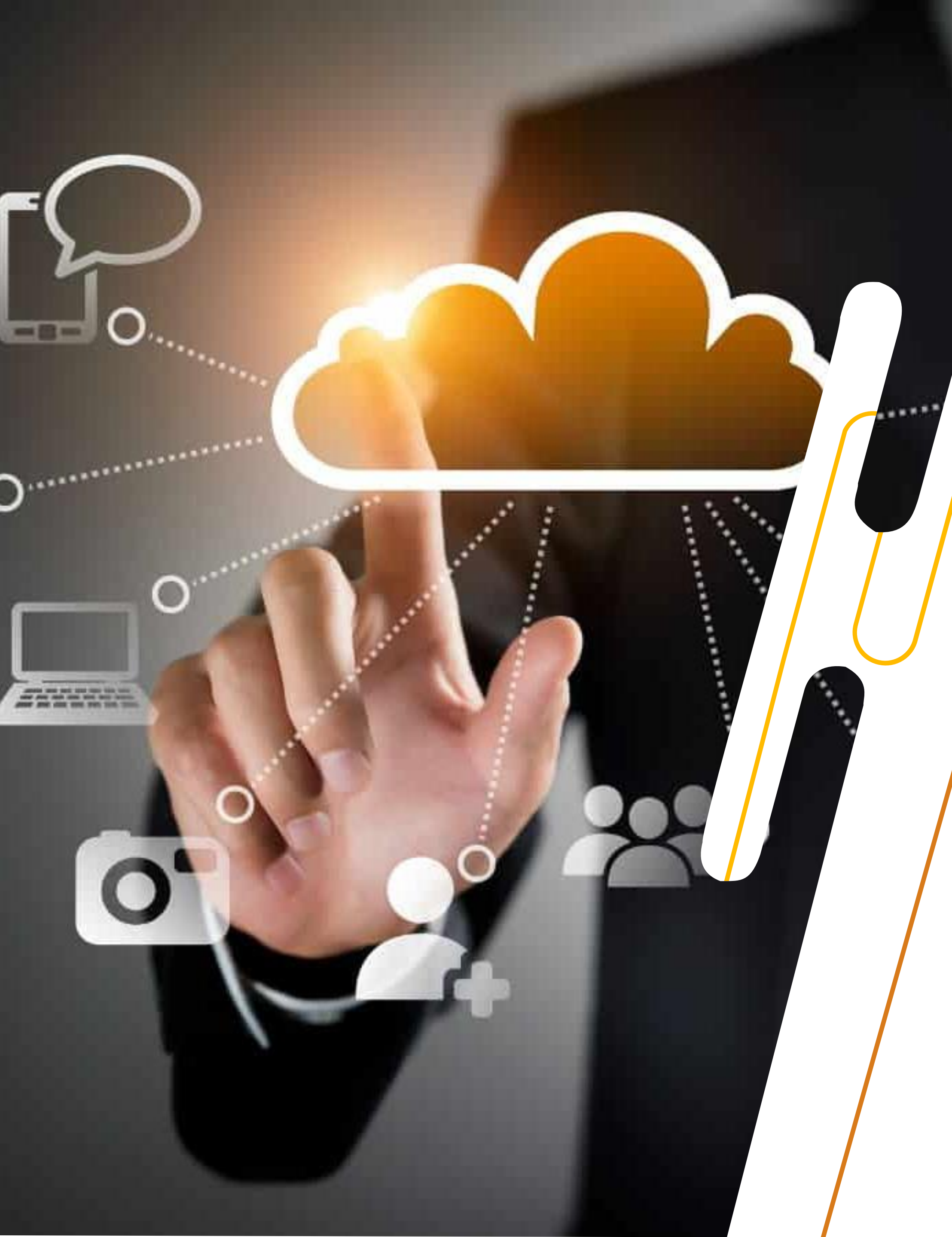


IT3061

Massive Data Processing and Cloud Computing

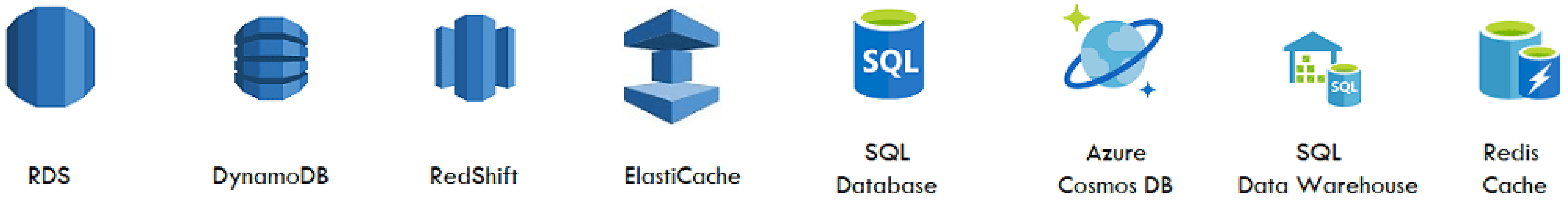
YEAR 3 SEMESTER 2
DATA SCIENCE SPECIALIZATION

Lecture Outline

01. Database as a Service concepts
02. Benefits and Drawbacks on DBaaS
03. DBaaS Types
04. How to choose DBaaS ?
05. Database Models, Workloads and caching
06. Data Warehouses
07. Data Lakes

Cloud Databases

- Database as a service (DBaaS) is a cloud computing managed service offering that provides access to a database without requiring the setup of physical hardware, the installation of software or the need to configure the database. Most maintenance and administrative tasks are handled by the service provider.
- The market for DBaaS and cloud databases is among the fastest-growing Software-as-a-Service (SaaS) markets, expected to grow to USD 320 billion by 2025



Cloud Databases

- Database hosting options are available for all database types, including NoSQL, MySQL, and PostgreSQL.

  MongoDB Atlas is one example of a NoSQL DBaaS service that is easily scalable.

- The DBaaS subscription includes everything required to operate a database in the cloud including
  - Database provisioning
  - Licenses
  - Support
  - Monitoring
  - Maintenance

Cloud Databases

- Developers can make use of cloud hosted APIs to build out new applications, accessing and manipulating data programmatically. Because of this, DBaaS shares many similarities with other SaaS subscription-based cloud offerings.
- Most database services offer web-based consoles, which the end user can use to provision and configure database instances.
- Database services consist of a database-manager component, which controls the underlying database instances using a service API. The service API is exposed to the end user, and permits users to perform maintenance and scaling operations on their database instances

Cloud Databases

- Underlying software stack typically includes the operating system, the database and third-party software used to manage the database. The service provider is responsible for installing, patching and updating the underlying software stack and ensuring the overall health and performance of the database.
- Scalability features differ between vendors - some offer auto-scaling, others enable the users to scale up using an API, but do not scale automatically. There is typically a commitment for a certain level of high availability (typically 99.9% or 99.99%)

Cloud Databases - Benefits

- Cost savings

  Pay a predictable periodic charge based on the resources you consume

- Scalability

  Quickly and easily provision additional storage and computing capacity at run and you can scale down your database cluster during non-peak usage times to save cost

- Simpler, less costly management

  The cloud provider manages everything (although you can choose to manage certain aspects yourself if you wish).

Cloud Databases - Benefits

- Rapid development and faster time-to-market

  Developers can help themselves to database capabilities and spin up and configure a database that's ready to integrate with their application in minutes

- Data and application security

  Cloud database providers typically offer enterprise grade security, including features like default encryption of data at rest and in-transit and integrated identity and access management controls. Some also meet specific regulatory compliance standards.

- Reduced risk

  DBaaS offerings from major cloud providers typically include a service-level agreement (SLA) guaranteeing a certain amount of uptime

Cloud Databases - Drawbacks

- Security and Data Privacy

  No visibility on how your data is being handled or managed as you do not have control or access to limit only people who can gain access

- Physical Control

  Whenever there are security and enhancements products that you wanted to push through and wanted to implement how your data is stored, accessed, or managed; there's no way you can implement it

Cloud Databases - Drawbacks

- Domain of Responsibility

  If data is lost or hardware goes rogue and affects your data, the question can be who has the responsibility of the data that was impacted

- Vendor Lock-In

How to choose the DBaaS ?

- Major cloud providers offer a wide array of DBaaS options, including relational database management systems, as well as non-relational or NoSQL databases, such as document and column stores.

- Finding the right DBaaS provider for your enterprise involves determining which database technologies will work best for your application and then, of course, ensuring that your provider supports that technology.

How to choose the DBaaS ?

- The first half of the process can be complex since there's no one-size-fits-all DBaaS that's optimal for use with all your applications. Trade-offs are always involved, and sometimes they can be subtle.
- Here are some specific factors you'll need to consider
  - Will a primary or auxiliary data store better suit my application?
  - Is the database's underlying architecture a good fit for my needs?
  - Does the database perform well during testing?

Database Models

Relational / SQL

- Highly structured table organization
- Rigidly-defined formats
- Dependencies among tables
- Enforce ACID (Atomicity, Consistency, Isolation, Durability)
- Reduces anomalies, enforces integrity
- Use SQL to access data
- Examples – MS SQL, MySQL, Oracle, PostgreSQL, Amazon RDS

Non-Relational / NoSQL

- Document oriented
- Large and complex queries
- Supports rapidly changing designs
- Examples – MongoDB, Cassandra, CosmosDB, Redis, CouchDB, Aurora

Database Workloads

Online Transaction Processing (OLTP)

- Focus is on operational data
- Transaction processing
- Small, simple ad-hoc queries
- Response in milliseconds
- Highly normalized

Online Analytical Processing (OLAP)

- Focus is on historical data
- Data analysis and reporting
- Large, complex queries
- Data warehouses
- Responses times from seconds to hours
- Typically denormalized

Database Caching

- Caching is a buffering technique that stores frequently requested data in temporary memory. Facilitates data access and reduces database workloads.
- Two popular caching systems
    - Redis
    - Memcached

Database Caching

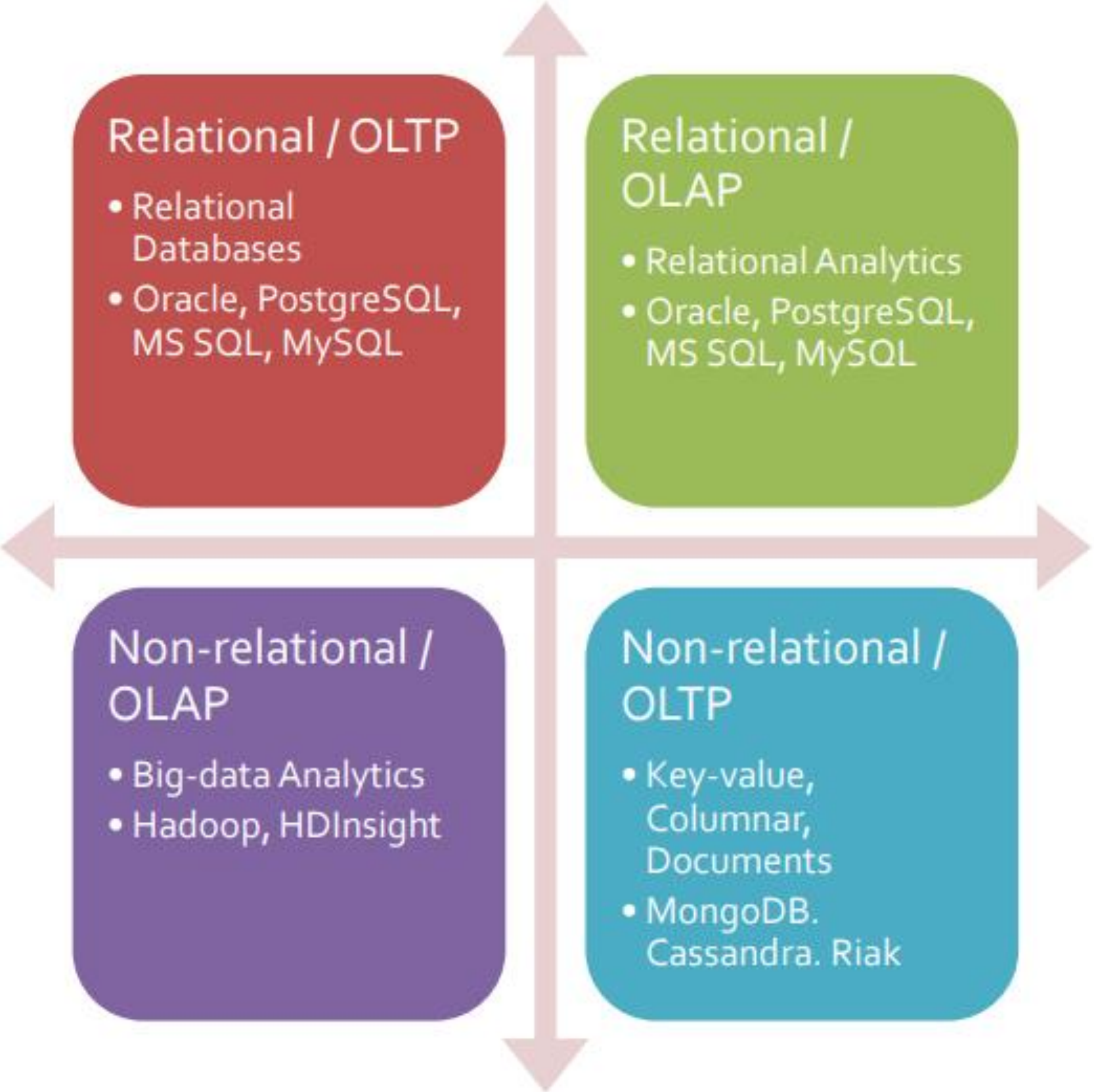
Redis

- Open source, in-memory, key-value data store
- Sub-millisecond response times
- Supports various data structures (strings, lists, sets etc.)
- Persistent – cache survives reboots
- Supports read replicas, atomic operations, backup/restore, HA

Memcached

- Open source, in-memory, object store
- Sub-millisecond response times
- Supports strings and objects
- Not persistent – cache does not survive reboots
- Supports scaling out, multithreading

Database Models and Workloads



Database Types and CSP offerings

	AWS	AZURE	GCP
RELATIONAL	■ Aurora (MySQL, PostgreSQL) ■ RDS (MySQL, MariaDB, Oracle, PostgreSQL, SQL Server) ■ Redshift	■ SQL Database ■ Azure Database (PostgreSQL, MySQL, MariaDB)	■ Cloud SQL(MySQL, PostgreSQL, SQL Server) ■ Cloud Spanner
KEY-VALUE	■ DynamoDB	■ Table Storage	■ Firestore
IN-MEMORY, CACHING	■ ElastiCache (Redis, Memcached)	■ Cache for Redis	■ Memorystore
DOCUMENT	■ DocumentDB (MongoDB compatible)	■ Cosmos DB	■ Firestore
WIDE COLUMN	■ Keyspaces (for Apache Cassandra)	■ Cosmos DB	■ Cloud Bigtable
GRAPH	■ Neptune	■ Cosmos DB	■ N/A**
TIME SERIES	■ Timestream	■ Time Series Insights ■ Cosmos DB	■ Cloud Bigtable
QUANTUM LEDGER	■ QLDB	■ N/A	■ N/A

Data Warehouse

- A data warehouse is a type of data management system designed to enable and support business intelligence (BI) activities, especially analytics.
- Data warehouses are intended for querying and analysis only and often contain large amounts of historical data.
- A repository for structured, filtered data that has already been processed for a specific purpose.
- Data in a data warehouse is typically derived from a wide variety of sources such as application log files and transaction applications.

Data Warehouse

- Two approaches

    ETL – Extract, Transform, Load (Source -> Staging -> Destination)

    ELT – Extract, Load, Transform (Source -> Destination)

Data Lake

- A data lake is a storage repository that holds a large amount of raw data in its native format until it is needed.
- While a hierarchical data warehouse stores data in files or folders, a data lake uses a flat architecture to store the data.

Data Warehouse VS Data Lake

Data Warehouse

Processed data

Data currently in use

Used by business professionals

Data Lake

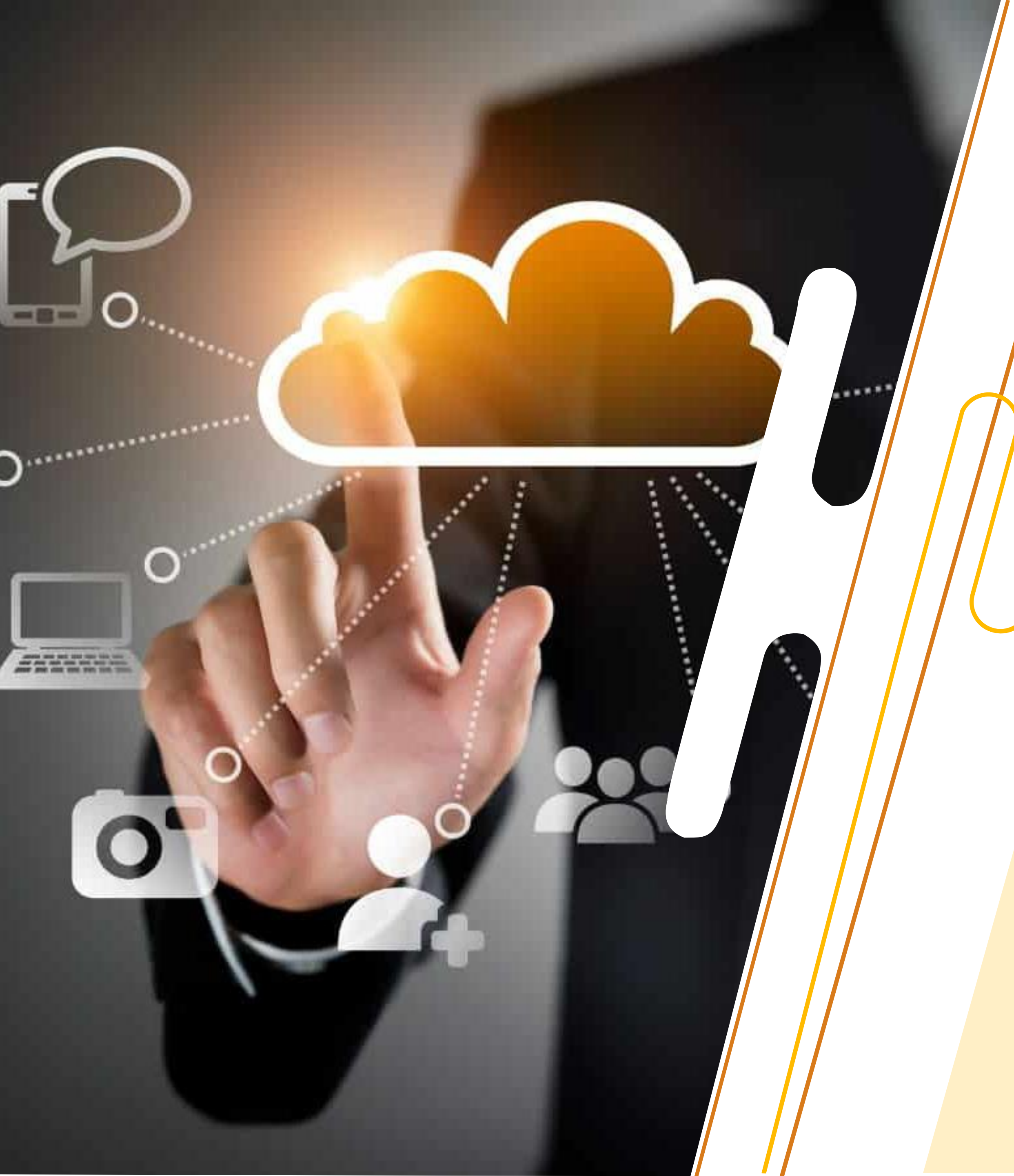
Raw data

Purpose of data not determined yet

Used by data scientists

Real Time Data Processing

- Real-time data processing is the quickest data processing technique that executes data in a short period of time and provides the most accurate output.
- The processing is done as the data is inputted, so it needs a continuous stream of input data in order to provide a continuous output.
- Also known as stream processing.



Demo

References

- AWS Academy Cloud Foundations
- AWS Documentation
- Azure Documentation

Next Session : Cloud Security